Lecture 6: Stochastic Convolutions, Additive Noise, and Weak Topology

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20.11.2025

§6 Continuity of Stochastic Convolutions

Remark 1 (Transcription note). This note is a transcription of the handwritten page. I keep the formulas close to the page, and I only add short bracketed comments when handwriting is unclear.

We start from the semilinear SPDE on a Hilbert space H:

$$\begin{cases} dX_t = \big(AX_t + B(t, X_t)\big)\, dt + C(t, X_t)\, dW_t, \\ X_0 = \xi. \end{cases} \qquad (X_t \in H)$$

Assume A generates a C_0-semigroup $(T_t)_{t\geq 0}$. The mild solution is written as

$$X_t = T_t\xi + \int_0^t T_{t-s}B(s, X_s)\, ds + \int_0^t T_{t-s}C(s, X_s)\, dW_s.$$

(*handwritten note under the first integral:* "drift".)
(*right-margin note near the second integral:* likely a comment about stochastic convolution / continuity; handwriting unclear.)

Existence of a continuous modification

Prop 2.2. Let $p > 1$ and

$$\Phi \in L^{2p}\big(\Omega_T, \mathcal{P}_T, P_T; L_2(U, H)\big).$$

(*small note near $\mathcal{P}_T$:* looks like "progressive proc"; likely referring to predictability/progressivity.)

Assume

$$E\left(\int_0^T \|\Phi(s) \circ \sqrt{Q}\|^{2p}_{L_2(U,H)}\, ds\right) < \infty.$$

Then there exists a constant C_T such that

$$E\left(\sup_{t\in[0,T]} \left\|\int_0^t T_{t-s}\Phi(s)\, dW_s\right\|^{2p}_H\right) \leq C_T\, E\left(\int_0^T \|\Phi(s) \circ \sqrt{Q}\|^{2p}_{L_2(U,H)}\, ds\right).$$

Moreover,

$$W_{A,\Phi}(t) := \int_0^t T_{t-s}\Phi(s)\, dW_s$$

has a continuous modification.

Remark 2. This is the standard maximal estimate for the stochastic convolution and the continuity statement obtained via the factorization method.

Proof (start). Will use the factorization method.

For $s < t$, use the β-integral identity

$$\int_s^t (t-r)^{\alpha-1}(r-s)^{-\alpha}\,dr = \frac{1}{c_\alpha} = \frac{\Gamma(\alpha)\Gamma(1-\alpha)}{\Gamma(1)} = \frac{\pi}{\sin(\pi\alpha)}.$$

[The handwritten constant may be c_α or C_α; the page is slightly ambiguous. Typically $\alpha \in (0,1)$.]

Remark 3 (Context for the identity). This is the Beta-function identity used in the factorization method. A common normalization is

$$c_\alpha = \frac{\sin(\pi\alpha)}{\pi},$$

so the integral equals $1/c_\alpha$.

Continuation of the proof (factorization step). Using the substitution

$$u := \frac{r-s}{t-s},$$

one gets the Beta integral in the normalized form

$$\int_0^1 (1-u)^{\alpha-1}u^{-\alpha}\,du.$$

Rewrite

$$W_{A,\Phi}(t) = \int_0^t T_{t-s}\Phi(s)\,dW_s$$

as

$$W_{A,\Phi}(t) = c_\alpha \int_0^t \left(\int_s^t (t-r)^{\alpha-1}(r-s)^{-\alpha}\,dr\right) T_{t-s}\Phi(s)\,dW_s.$$

(*handwritten note:* "semigroup", indicating the split $T_{t-s} = T_{t-r}T_{r-s}$.)

Applying the stochastic Fubini theorem (and changing the order of integration) gives

$$\begin{aligned} W_{A,\Phi}(t) &= c_\alpha \int_0^t \int_0^r (t-r)^{\alpha-1}(r-s)^{-\alpha} T_{t-s}\Phi(s)\,dW_s\,dr \\ &= c_\alpha \int_0^t (t-r)^{\alpha-1} T_{t-r}\left(\int_0^r (r-s)^{-\alpha} T_{r-s}\Phi(s)\,dW_s\right) dr. \end{aligned}$$

Define

$$y(r) := \int_0^r (r-s)^{-\alpha} T_{r-s}\Phi(s)\,dW_s,$$

so that

$$W_{A,\Phi}(t) = c_\alpha \int_0^t (t-r)^{\alpha-1} T_{t-r} y(r)\,dr.$$

(*handwritten margin note:* this is a deterministic convolution in t, hence suggests continuity once integrability is checked.)

Remark 4 (Choice of α). The handwritten line appears to indicate the usual condition for Hölder integrability: choose α so that $\frac{1}{2p} < \alpha < \frac{1}{2}$ (so that $\frac{1}{2\alpha} > 1$). In particular, the integrability condition later becomes

$$p > \frac{1}{2\alpha} > 1.$$

Then, for $t \in [0,T]$,

$$\|W_{A,\Phi}(t)\|_H^{2p} \le c_\alpha^{2p} \left(\int_0^t (t-r)^{\alpha-1} \|T_{t-r} y(r)\|_H \, dr \right)^{2p}.$$

(*handwritten note near this step:* "Bochner int".)

By Hölder's inequality,

$$\begin{aligned}
\|W_{A,\Phi}(t)\|_H^{2p} &\le c_\alpha^{2p} \left(\int_0^t (t-r)^{(\alpha-1)\frac{2p}{2p-1}} \, dr \right)^{2p-1} \int_0^t \|T_{t-r} y(r)\|_H^{2p} \, dr \\
&\le c_\alpha^{2p} \left(\frac{T^{1+(\alpha-1)\frac{2p}{2p-1}}}{1+(\alpha-1)\frac{2p}{2p-1}} \right)^{2p-1} \int_0^t \|T_{t-r} y(r)\|_H^{2p} \, dr \\
&\le C_{\alpha,p,T} \int_0^t \|T_{t-r} y(r)\|_H^{2p} \, dr.
\end{aligned}$$

(*handwritten side note near the denominator/exponent line:* "have to check integrability".)

Remark 5 (What integrability is being checked). In the Hölder step one needs

$$\int_0^t (t-r)^{(\alpha-1)\frac{2p}{2p-1}} \, dr < \infty,$$

which holds precisely when $(\alpha-1)\frac{2p}{2p-1} > -1$, i.e. $p > \frac{1}{2\alpha}$.

Estimating the auxiliary process $y(r)$. The integrability condition in the Hölder step is

$$(\alpha-1)\frac{2p}{2p-1} > -1 \qquad \Longleftrightarrow \qquad p > \frac{1}{2\alpha},$$

which is consistent with the earlier choice $\alpha > \frac{1}{2p}$.

The BDG inequality (generalization of Itô isometry to higher moments) yields, for each $r \in [0,T]$,

$$E\Big(\|y(r)\|_H^{2p} \Big) \le C_p \, E\left[\left(\int_0^r (r-s)^{-2\alpha} \big\| T_{r-s} \Phi(s) \circ \sqrt{Q} \big\|_{L_2(U,H)}^2 \, ds \right)^p \right].$$

(*handwritten margin note:* "BDG inequality implies (generalization of Itô isometry for any exponent p)".)

Using the semigroup bound (absorbed into constants), we obtain

$$\int_0^T E\Big(\|y(r)\|_H^{2p} \Big) \, dr \le C_p \, E\left[\int_0^T \left(\int_0^r (r-s)^{-2\alpha} \|\Phi(s) \circ \sqrt{Q}\|_{L_2(U,H)}^2 \, ds \right)^p dr \right].$$

By Hölder's inequality in the inner integral,

$$\left(\int_0^r (r-s)^{-2\alpha} \|\Phi(s) \circ \sqrt{Q}\|_{L_2(U,H)}^2 \, ds \right)^p \le \left(\int_0^r (r-s)^{-2\alpha} \, ds \right)^{p-1} \int_0^r (r-s)^{-2\alpha} \|\Phi(s) \circ \sqrt{Q}\|_{L_2(U,H)}^{2p} \, ds.$$

Since $\alpha < \frac{1}{2}$ (the standard factorization choice), we have

$$\int_0^r (r-s)^{-2\alpha} \, ds \;=\; \int_0^r u^{-2\alpha} \, du \;\le\; \int_0^T u^{-2\alpha} \, du \le \int_0^T r^{-2\alpha} \, dr = \frac{T^{1-2\alpha}}{1-2\alpha}.$$

Hence, by Fubini/Tonelli,

$$\begin{aligned}
\int_0^T E\Big(\|y(r)\|_H^{2p}\Big)\,dr &\le C_{p,\alpha,T}\,E\Big[\int_0^T\int_0^r (r-s)^{-2\alpha}\|\Phi(s)\circ\sqrt{Q}\|_{L_2(U,H)}^{2p}\,ds\,dr\Big]\\
&= C_{p,\alpha,T}\,E\Big[\int_0^T\Big(\int_s^T (r-s)^{-2\alpha}\,dr\Big)\,\|\Phi(s)\circ\sqrt{Q}\|_{L_2(U,H)}^{2p}\,ds\Big]\\
&\le C_{p,\alpha,T}\,E\Big[\int_0^T\Big(\int_0^T u^{-2\alpha}\,du\Big)\,\|\Phi(s)\circ\sqrt{Q}\|_{L_2(U,H)}^{2p}\,ds\Big]\\
&\le C_{p,\alpha,T}\,E\Big[\int_0^T \|\Phi(s)\circ\sqrt{Q}\|_{L_2(U,H)}^{2p}\,ds\Big] < \infty.
\end{aligned}$$

Therefore,

$$\int_0^T \|y(r)\|_H^{2p}\,dr < \infty \qquad \text{a.s.}$$

In other words, $y(\cdot,\omega)\in L^{2p}([0,T];H)$ for a.e. $\omega\in\Omega$.

Main implication (continuity of the factorized term). The handwritten note states that the deterministic convolution operator

$$g \longmapsto \Big[t \longmapsto \int_0^t (t-r)^{\alpha-1}T_{t-r}g(r)\,dr\Big]$$

sends

$$L^{2p}([0,T];H) \quad \text{into} \quad C([0,T];H),$$

which gives continuity of $t\mapsto W_{A,\Phi}(t)$ since we already have the factorization identity

$$W_{A,\Phi}(t) = c_\alpha \int_0^t (t-r)^{\alpha-1}T_{t-r}y(r)\,dr.$$

(*handwritten side note near this line appears to say:* "absolutely contin[uous]" / "continuous"; exact wording is unclear.)

Remark 6 (Deterministic mapping property used above). Let $(T_t)_{t\ge0}$ be a C_0-semigroup on H. Fix $q>1$ and $\alpha\in(0,1)$ with $\alpha>\frac{1}{q}$. For $g\in L^q([0,T];H)$ define

$$(\mathcal{K}g)(t) := \int_0^t (t-r)^{\alpha-1}T_{t-r}g(r)\,dr.$$

Then $\mathcal{K}g\in C([0,T];H)$ and

$$\sup_{t\in[0,T]} \|(\mathcal{K}g)(t)\|_H \le C_{\alpha,q,T}\,\|g\|_{L^q([0,T];H)}.$$

Sketch: by Hölder,

$$\|(\mathcal{K}g)(t)\|_H \le \Big(\int_0^t (t-r)^{(\alpha-1)q'}\,dr\Big)^{1/q'}\|g\|_{L^q([0,t];H)}, \qquad q'=\frac{q}{q-1},$$

and $\int_0^t (t-r)^{(\alpha-1)q'}dr<\infty$ iff $(\alpha-1)q'>-1$, i.e. $\alpha>\frac{1}{q}$. Continuity in t follows from strong continuity of T_t and dominated convergence using this bound. *More explicitly:* if $t_n\to t$, then for a.e. $r\in(0,t)$ one has $(t_n-r)^{\alpha-1}T_{t_n-r}g(r)\to(t-r)^{\alpha-1}T_{t-r}g(r)$, and for n large,

$$\|(t_n-r)^{\alpha-1}T_{t_n-r}g(r)\|_H \le C_T\,(t-r)^{\alpha-1}\|g(r)\|_H,$$

where $C_T := \sup_{0\le\tau\le T}\|T_\tau\| < \infty$. The right-hand side is in $L^1(0,t)$ by Hölder (since $(t-r)^{\alpha-1} \in L^{q'}(0,t)$ and $g \in L^q(0,t;H)$), hence dominated convergence applies. *That is,* dominated convergence yields

$$\int_0^t (t_n - r)^{\alpha-1} T_{t_n-r} g(r)\,dr \longrightarrow \int_0^t (t-r)^{\alpha-1} T_{t-r} g(r)\,dr \qquad \text{in } H.$$

The remaining tail term satisfies, for $t_n \ge t$,

$$\left\| \int_t^{t_n} (t_n - r)^{\alpha-1} T_{t_n-r} g(r)\,dr \right\|_H \le C_T \left(\int_t^{t_n} (t_n - r)^{(\alpha-1)q'}\,dr \right)^{1/q'} \|g\|_{L^q([t,t_n];H)} \to 0,$$

so $(\mathcal{K}g)(t_n) \to (\mathcal{K}g)(t)$.

Remark 7. The page cites this deterministic mapping property as a known fact (with a note pointing to lecture notes / theory of evolution equations). This is the final step turning the factorization representation into existence of a continuous modification.

Remark: Additive Noise

Consider the stochastic differential equation with *additive noise*

$$dX_t = \big(AX_t + B(X_t)\big)\,dt + C\,dW_t,$$

where C does not depend on the state X_t.

Let

$$W_{A,C}(t) := \int_0^t T_{t-s} C\,dW_s.$$

Then one can decompose

$$X_t = Y_t + W_{A,C}(t).$$

Writing the mild formulation for X and substituting $X_s = Y_s + W_{A,C}(s)$ gives

$$\begin{aligned} X_t &= T_t\xi + \int_0^t T_{t-s} B(X_s)\,ds + W_{A,C}(t) \\ &= T_t\xi + \int_0^t T_{t-s} B\big(Y_s + W_{A,C}(s)\big)\,ds + W_{A,C}(t). \end{aligned}$$

Since $X_t = Y_t + W_{A,C}(t)$ by definition, the left-hand side is $Y_t + W_{A,C}(t)$, so subtracting $W_{A,C}(t)$ from both sides yields

$$Y_t = T_t\xi + \int_0^t T_{t-s} B\big(Y_s + W_{A,C}(s)\big)\,ds,$$

which is the mild form of the (random) deterministic evolution equation

$$\begin{cases} dY_t = AY_t\,dt + B\big(Y_t + W_{A,C}(t)\big)\,dt, \\ Y_0 = \xi. \end{cases}$$

Here "(random) deterministic" means: for each fixed outcome $\omega \in \Omega$ the path $t \mapsto W_{A,C}(t,\omega)$ is given, so $Y(\cdot,\omega)$ solves an ordinary deterministic evolution equation in time, but with coefficients/forcing depending on ω.

Remark 8. This reduces the additive-noise SPDE to a deterministic (but random-coefficient) evolution equation for Y. The handwritten margin note appears to say this "accords" with the stochastic convolution decomposition.

Interlude (Appendix B): Weak Topology

Let E be a real Banach space.

Definition (weak convergence). A sequence $(x_n) \subset E$ converges *weakly* to $x \in E$ if

$$\langle f, x_n \rangle_{E',E} \longrightarrow \langle f, x \rangle_{E',E} \qquad \text{for all } f \in E'.$$

Equivalently,

$$f(x_n) \to f(x) \qquad \text{for all } f \in E',$$

where E' denotes the dual space of bounded linear functionals on E.

We write

$$x_n \rightharpoonup x$$

in this case.

Remark 9. The notation on the page uses the dual pairing $\langle f, x \rangle$ and also writes it as $f(x)$.

Remark. (Strong) convergence with respect to the norm implies weak convergence. Indeed, if

$$\|x_n - x\|_E \to 0,$$

then for every $f \in E'$,

$$\big|\langle f, x_n \rangle - \langle f, x \rangle\big| = \big|\langle f, x_n - x \rangle\big| \le \|f\|_{E'} \, \|x_n - x\|_E \longrightarrow 0.$$

(*handwritten note at the bottom is hard to read; it appears to reference the standard duality estimate / Cauchy–Schwarz-type bound in the dual pairing.*)

Remark (converse is false). The converse is not true in general: weak convergence does *not* imply norm convergence.

Take the canonical basis (e_n) in ℓ^2, e.g.

$$e_n = (\delta_{kn})_{k \in \mathbb{N}} \in \ell^2.$$

Equivalently, $(e_n)_k = \delta_{kn}$ for each $k \in \mathbb{N}$, i.e. $e_n = (0, \dots, 0, 1, 0, \dots)$ with the 1 in the n-th position. Then

$$\|e_n\|_{\ell^2} = 1 \qquad \text{for all } n,$$

so (e_n) does not converge to 0 in norm. However,

$$e_n \rightharpoonup 0 \quad \text{in } \ell^2.$$

Indeed, for every $f \in (\ell^2)'$ (identified with an element of ℓ^2 via the Riesz representation theorem),

$$\langle f, e_n \rangle_{\ell^2} = f_n \longrightarrow 0.$$

Indeed, if $f = (f_k)_{k \in \mathbb{N}} \in \ell^2$, then $\sum_{k=1}^{\infty} |f_k|^2 < \infty$, hence necessarily $|f_n| \to 0$ (otherwise the series could not converge). (*handwritten note near this line:* "converges weakly".)

Hahn–Banach theorem (extension form, as used here). Let E be a real normed space, let $E_0 \subset E$ be a linear subspace, and let

$$\ell\colon E_0 \to \mathbb{R}$$

be linear and continuous (bounded with respect to the norm of E). Then there exists an extension

$$\widetilde{\ell} \in E'$$

such that

$$\widetilde{\ell}\restriction_{E_0} = \ell \qquad \text{and} \qquad \|\widetilde{\ell}\|_{E'} = \|\ell\|_{E_0'}.$$

Remark (uniqueness of weak limit). The weak limit is uniquely determined. Equivalently,

$$\langle f, x\rangle = 0 \quad \text{for all } f \in E' \qquad \Longrightarrow \qquad x = 0.$$

This follows from Hahn–Banach: for every $x \in E$ with $x \neq 0$, there exists $f \in E'$ such that $f(x) \neq 0$. More explicitly: if $x \in E$ with $x \neq 0$, consider the 1D subspace $M := \operatorname{span}\{x\}$ and define $g\colon M \to \mathbb{K}$ by

$$g(\alpha x) := \alpha \|x\|_E,$$

so g is linear and $\|g\|_{M'} = 1$. Hahn–Banach extends g to some $f \in E'$ with $\|f\|_{E'} = 1$. In particular, $f(x) = g(x) = \|x\|_E \neq 0$. Hence for every nonzero x there exists $f \in E'$ with $f(x) \neq 0$.

Remark 10 (Motivation note from the page). The handwritten page sketches the standard normalization argument: assume w.l.o.g. that

$$|\ell(u)| \leq \|u\| \qquad (u \in E_0),$$

then Hahn–Banach gives an extension $\widetilde{\ell} \in E'$ with

$$\widetilde{\ell}|_{E_0} = \ell \qquad \text{and} \qquad \|\widetilde{\ell}\|_{E'} \leq 1.$$

Proposition (basic consequences of weak convergence). If $x_n \rightharpoonup x$ in E, then:

1. the sequence (x_n) is bounded in E;
2. $\|x\|_E \leq \liminf_{n\to\infty} \|x_n\|_E$.

Remark 11 (Proof idea recorded on the page). For (ii), choose (by Hahn–Banach) $g \in E'$ with $\|g\|_{E'} = 1$ and $g(x) = \|x\|_E$ (for $x \neq 0$; the case $x = 0$ is trivial). Then

$$\|x\|_E = |g(x)| = \lim_{n\to\infty} |g(x_n)| \;=\; \liminf_{n\to\infty} |g(x_n)| \leq \liminf_{n\to\infty} \|g\|_{E'} \|x_n\|_E = \liminf_{n\to\infty} \|x_n\|_E.$$

The boundedness of (x_n) is standard (e.g. by the uniform boundedness principle applied to the functionals $f \mapsto f(x_n)$ on E').

Remark 12 (Why weak convergence implies boundedness). If $x_n \rightharpoonup x$ in E, then for each $f \in E'$ the scalar sequence $f(x_n) \to f(x)$ is bounded, i.e.

$$\sup_{n\in\mathbb{N}} |f(x_n)| < \infty.$$

Define $T_n \in \mathcal{L}(E', \mathbb{K})$ by $T_n(f) := f(x_n)$. For each fixed $f \in E'$, weak convergence gives $T_n(f) = f(x_n) \to f(x)$, hence $\sup_n |T_n(f)| < \infty$. Thus $\{T_n\}$ is pointwise bounded on E', and by the Banach–Steinhaus theorem,

$$\sup_n \|T_n\|_{\mathcal{L}(E',\mathbb{K})} < \infty.$$

But $\|T_n\| = \sup_{\|f\|_{E'}=1} |f(x_n)| = \|x_n\|_E$ (via the canonical embedding $E \hookrightarrow E''$), hence $\sup_n \|x_n\|_E < \infty$.

Proposition (Hilbert space criterion). In a Hilbert space H, weak convergence together with convergence of norms implies strong convergence:

$$x_n \rightharpoonup x \quad \text{and} \quad \|x_n\|_H \to \|x\|_H \qquad \Longrightarrow \qquad x_n \to x \text{ in } H.$$

Proof (start, as on the page).

$$\|x_n - x\|_H^2 = \|x_n\|_H^2 - 2\langle x_n, x\rangle_H + \|x\|_H^2.$$

Since $x_n \rightharpoonup x$, for every fixed $y \in H$ we have $\langle x_n, y\rangle_H \to \langle x, y\rangle_H$. In particular (taking $y = x$),

$$\langle x_n, x\rangle_H \longrightarrow \langle x, x\rangle_H = \|x\|_H^2.$$

Moreover, $\|x_n\|_H \to \|x\|_H$ implies $\|x_n\|_H^2 \to \|x\|_H^2$. Therefore,

$$\|x_n - x\|_H^2 \longrightarrow \|x\|_H^2 - 2\|x\|_H^2 + \|x\|_H^2 = 0,$$

so $\|x_n - x\|_H \to 0$, i.e. $x_n \to x$ strongly.
(This is exactly the limit computation written at the top of the next page:

$$\|x\|_H^2 - 2\langle x, x\rangle_H + \|x\|_H^2 = 0.$$

)

Theorem (Banach–Alaoglu). This is an extension of the Bolzano–Weierstrass theorem to dual spaces (weak* compactness).

The closed unit ball (more generally, every norm-bounded subset) of E' is compact in the weak* topology

$$\sigma(E', E).$$

Recall that the weak* topology $\sigma(E', E)$ is the coarsest topology on E' (i.e. the one with the fewest open sets / weakest notion of convergence) making all evaluation maps

$$E' \ni f \longmapsto \langle f, x\rangle \in \mathbb{K} \qquad (x \in E)$$

continuous. Equivalently, $f_n \overset{*}{\rightharpoonup} f$ means pointwise convergence on E:

$$\langle f_n, x\rangle \to \langle f, x\rangle \quad \text{for all } x \in E.$$

This is typically much weaker than norm convergence $f_n \to f$ in E'. If the predual E is separable, then on bounded subsets this weak* compactness is sequential (i.e. one gets weak* convergent subsequences).

Remark 13 (Handwritten formulation on the page). The page phrases this as: any bounded sequence $(f_n) \subset E'$ has a weak* convergent subsequence (when the relevant separability assumption is available), meaning

$$f_{n_k} \overset{*}{\rightharpoonup} f \quad \text{in } E' \qquad \Longleftrightarrow \qquad \langle f_{n_k}, x\rangle \to \langle f, x\rangle \text{ for all } x \in E.$$

Remark 14 (Unit ball in the reflexive case). Banach–Alaoglu says that the closed unit ball of E' is compact in $\sigma(E', E)$ (weak*). Applying Banach–Alaoglu to E'' (the dual of E') gives that the closed unit ball of E'' is compact in $\sigma(E'', E')$. If E is reflexive, we identify $E \simeq E''$ via the canonical embedding J, and the weak* topology $\sigma(E'', E')$ restricted to $J(E)$ coincides with the weak topology $\sigma(E, E')$ on E. Hence the closed unit ball of E is weakly compact (and therefore relatively weakly compact).

Reflexive-space consequence (in particular for Hilbert spaces). If E is reflexive (e.g. a Hilbert space), then bounded subsets of E are relatively weakly compact. If in addition E is separable, bounded sequences admit weakly convergent subsequences.

Remark 15 (What reflexive means / why it helps). A Banach space E is called *reflexive* if the canonical embedding $J\colon E \to E''$ is surjective (so E can be identified with E''). Here the canonical embedding is

$$J(x)(f) := f(x), \qquad x \in E,\ f \in E',$$

i.e. $J(x) \in E''$ is the functional on E' given by evaluation at x. Banach–Alaoglu gives weak* compactness of bounded sets in E'' (as the dual of E'). If E is reflexive, bounded subsets of E become weak* bounded subsets of E'' and hence are relatively compact for the weak topology on E. Hilbert spaces are reflexive, so bounded sequences in Hilbert spaces always have weakly convergent subsequences (and in separable Hilbert spaces one can choose subsequences sequentially).

Remark 16 (Meaning of "weakly compact"). A set $K \subset E$ is *weakly compact* if it is compact with respect to the weak topology on E (not necessarily compact in the norm topology). If E is separable, weak compactness on bounded sets is sequential, i.e. it can be tested by existence of weakly convergent subsequences.

Remark 17 (Meaning of "relatively weakly compact"). A set $S \subset E$ is *relatively weakly compact* if its closure in the weak topology of E is weakly compact. In particular, if E is separable, relative weak compactness implies: every bounded sequence in S has a weakly convergent subsequence with limit in the weak closure of S.

Remark (weak convergence of measures). For finite measures (or probability measures) μ_n, μ on Ω, one writes

$$\mu_n \Rightarrow \mu$$

for weak convergence, and the page notes the functional-analytic characterization

$$\mu_n \Rightarrow \mu \qquad \Longleftrightarrow \qquad \int_\Omega f\, d\mu_n \to \int_\Omega f\, d\mu \quad \text{for all test functions } f \text{ (typically } f \in C_b(\Omega) \text{ for probability measures).}$$

(*handwritten note:* "functional analytic sense" / "weak conv" near the integral criterion.)

Here *finite measure* means $\mu(\Omega) < \infty$. Probability measures are the special case $\mu(\Omega) = 1$. In the probability-measure case, $\mu_n \Rightarrow \mu$ is also called *weak convergence of probabilities* or *convergence in distribution* (convergence in law). Equivalently, if X_n, X are random variables with $\mathcal{L}(X_n) = \mu_n$ and $\mathcal{L}(X) = \mu$, then

$$X_n \xrightarrow{d} X \qquad \Longleftrightarrow \qquad \mu_n \Rightarrow \mu.$$

Remark 18 (About the test-function class). The page appears to mention a space like $C(\Omega)$ or $C_0(\Omega)$ (the handwriting is slightly unclear). Standard versions are:

$$f \in C_b(\Omega) \quad \text{for weak convergence of probability measures,}$$

or

$$f \in C_0(\Omega) \quad \text{in the Riesz–Markov / Radon-measure duality setting.}$$

Here $C_b(\Omega)$ denotes the space of bounded continuous functions on Ω, and (for locally compact Hausdorff Ω) $C_0(\Omega)$ denotes the space of continuous functions vanishing at infinity.

Riesz–Radon (Riesz representation) reminder. The final line on the page records the duality identification (in an abbreviated form):

$$C(\Omega)' \cong \{\text{finite signed measures on } \Omega\},$$

In words: $C(\Omega)'$ is isomorphic to the space of finite signed measures on Ω. where *finite signed* means $|\mu|(\Omega) < \infty$. In words: every continuous linear functional on $C(\Omega)$ is uniquely obtained by integrating a continuous function against a finite signed measure, and conversely every finite signed measure gives such a functional. For compact Hausdorff Ω this gives the dual of $C(\Omega)$, while for locally compact Hausdorff Ω one uses $C_0(\Omega)$ and obtains finite signed Radon measures.

Remark 19 (Terminology).

- A *signed measure* μ on Ω is a countably additive set function (taking values in $\mathbb{R}$ or $\mathbb{C}$). It is *finite* if its total variation satisfies $|\mu|(\Omega) < \infty$.
- A (signed) *Radon measure* on a locally compact Hausdorff space Ω is a Borel measure that is finite on compact sets and regular (measurable sets can be approximated from inside by compacts and from outside by opens).
- *Locally compact Hausdorff* means: Ω is Hausdorff, and every point has a neighborhood whose closure is compact.